

# The Field Trial

Experimental design, heterogeneous effects and identification

Companion to the standalone HTML game

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**Abstract.** A public programme offers training, transport support, or both. Players commission randomised pilots to separate the effects of the two programmes and the extra effect of combining them, allowing responses to differ across population groups. Each partner recruits a known mixture of groups but releases only a pooled income contrast. Programme availability, delivery prices and one-off opening fees constrain experimental design. A numerical report must be uniquely supported by the purchased evidence; a non-identification report must exhibit a nonzero change in effects that every permitted pilot would miss. False or unsupported reports score zero. Correct reports are scored against an exactly computed minimum-cost design. Four missions distinguish confounding between programme components, heterogeneous responses, costly recruitment and a failure of identification despite representation of every group.

## 1 The player’s task

The policy question is concrete: *what does training do on its own, what does transport do on its own, and how much more or less do they achieve together?* The population has four observable groups: lower or higher qualifications, crossed with living near or far from jobs. Some missions allow eight distinct effects; another allows twelve.

The interface presents seven recruitment partners. A coloured two-by-two picture shows each partner’s population mix. The player chooses two programme versions: neither programme, training only, transport only, or both. One is the reference; the reported result is the other version’s average income minus the reference’s average income. Some partners cannot offer every version.

Players can add and remove pilots from a plan without spending. Running a pilot pays for two fresh trial arms and, if necessary, opens its recruitment partner. Opening fees are charged only once. The results notebook preserves every observed contrast and any free results supplied at the start. There is no timer and no reward for mental arithmetic.

The final report has two alternatives. **Give the effects** requests all entries in the effect table. A calculator can supply one model fitting the observed results, but fitting is not a test of uniqueness. **Show why impossible** requests changes to those entries that cancel in every permitted comparison. At least one change must be nonzero. A confirmation seals the report and ends the round; the player then receives an explanation and the benchmark design.

**A deliberately precise information environment.** The partners report pooled differences, not outcomes linked to demographic group labels. Their recruitment mixes are fixed. Each contract buys a separate two-arm pilot. These restrictions are stated before play: the game is not claiming that a researcher with unrestricted individual data and a freely designed factorial experiment would face the same problem. Exact population results remove sampling uncertainty so that the task concerns identification and research design.

## 2 Economic model and the identification criterion

Index groups by  $g \in \{LN, LF, HN, HF\}$ , with  $L, H$  denoting qualifications and  $N, F$  distance. Let  $T, R \in \{0, 1\}$  indicate training and transport. Within pilot  $e$ , the group mean potential outcome is

$$\mu_{eg}(T, R) = m_{eg} + a_g T + b_g R + c_g TR.$$

The three effects are defined by the four group mean outcomes:

$$\begin{aligned} a_g &= \mu_{eg}(1, 0) - \mu_{eg}(0, 0), \\ b_g &= \mu_{eg}(0, 1) - \mu_{eg}(0, 0), \\ c_g &= \mu_{eg}(1, 1) - \mu_{eg}(1, 0) - \mu_{eg}(0, 1) + \mu_{eg}(0, 0). \end{aligned}$$

Thus the combined gain is  $a_g + b_g + c_g$ . A negative  $c_g$  means that the joint gain falls short of the sum of the separate gains. The expression is saturated for two binary programmes at the level of group means; it does not require additive effects. It decomposes programme contrasts, not latent psychological or behavioural mediation mechanisms.

Baseline incomes  $m_{eg}$  may differ between pilots. Effects  $a_g, b_g, c_g$  are stable across pilots and partners for a given group. Randomisation gives the same group proportions in the two arms, so the baseline cancels within a pilot. Full delivery, no spillovers and no carryover are assumed. Cross-pilot differences in income levels are not available as causal comparisons.

For partner  $s$ , let  $p_{sg}$  be the known recruitment proportion, with  $\sum_g p_{sg} = 1$ . A pilot compares versions  $u = (T_u, R_u)$  and  $v = (T_v, R_v)$ . Its only outcome is

$$y_e = \sum_g p_{sg} \{a_g \Delta T + b_g \Delta R + c_g \Delta(TR)\}, \quad \Delta T = T_v - T_u,$$

with analogous definitions for the other differences. Both arms have mix  $p_s$ .

**Two model classes.** The eight-effect missions impose

$$a_{LN} = a_{LF} = a_L, \quad a_{HN} = a_{HF} = a_H, \quad b_{LN} = b_{HN} = b_N, \quad b_{LF} = b_{HF} = b_F,$$

while leaving all four interactions unrestricted. Their parameter vector is

$$\theta = (a_L, a_H, b_N, b_F, c_{LN}, c_{LF}, c_{HN}, c_{HF})' \in \mathbb{R}^8.$$

The twelve-effect mission instead uses all four  $a_g$ , all four  $b_g$  and all four  $c_g$ , in that order, with  $\theta \in \mathbb{R}^{12}$ . These are stated model restrictions, not conclusions obtained by testing equality in the pilots.

Every observation can be written  $y_e = x'_e \theta$ , where the known row  $x'_e$  follows from the programme contrast and recruitment mix. Stacking the recorded rows gives  $y = X\theta$ .

**Proposition 1.** *For the unrestricted real parameter space of dimension  $d$ , the full requested vector is identified by the recorded results if and only if  $\text{rank}(X) = d$ .*

*Proof.* Full column rank gives at most one solution, and the true vector supplies one. Otherwise there is a nonzero  $\delta$  with  $X\delta = 0$ . Every  $\theta + t\delta$ ,  $t \in \mathbb{R}$ , fits the same evidence. No finite menu, sign restriction or rounding grid excludes these alternatives.  $\square$

### 3 Useful strategies and misleading shortcuts

**Separate programme contrasts before interpreting mechanisms.** A trial of both programmes against neither measures a mixture of  $a$ ,  $b$  and  $c$ . It cannot attribute the gain to training or transport. Adding training to existing transport measures  $a + c$ , not  $a$ . Adding transport to existing training measures  $b + c$ . These comparisons can be valuable, but their interpretation depends on the other available evidence.

For a common recruitment mix  $p$ , three appropriately chosen contrasts recover the three *pooled* components. For example, with  $0, T, R, B$  denoting neither, training, transport and both,

$$y_{0B} - y_{0T} - y_{0R} = \sum_g p_g c_g.$$

This isolates an average interaction. It does not separate its four group effects. Nor do three comparisons at one site identify the entire heterogeneous policy response.

**Vary recruitment as well as treatment.** In the twelve-effect model, obtaining three independent programme contrasts at each of four partners with linearly independent recruitment vectors is sufficient. This is a useful starting design, but it need not be the cheapest design. Partners with restricted programme menus may provide an inexpensive part of the required variation. In the eight-effect model, the shared-effect assumptions allow some information to be pooled across groups, making complete local factorial designs still less necessary.

**Count independent information, not purchases.** At the same partner,

$$x_{0B} = x_{0T} + x_{TB}, \quad y_{0B} = y_{0T} + y_{TB}.$$

Once two sides of this triangle have been measured, the third supplies no new restriction. Different partners can also supply dependent information when their recruitment mixes are related. A distinct-looking pilot can therefore be redundant even if it is randomised and reports a new numerical contrast. The game prevents exact duplicate purchases, but it allows these subtler mistakes.

**Plan openings jointly with comparisons.** A low setup fee can conceal expensive delivery, while an initially expensive partner may supply several cheap useful comparisons. Selecting the cheapest currently independent pilot, including its immediate opening fee, is not generally optimal. The second mission makes that policy exceed the budget. An expert must consider the study as a whole, including which partners will be used repeatedly.

**Distinguish failure so far from impossibility.** Let  $X_{\text{all}}$  contain every permitted comparison, whether purchased or not. A nonzero  $\delta$  with  $X\delta = 0$  establishes ambiguity in the current notebook. It establishes structural non-identification only if

$$X_{\text{all}}\delta = 0.$$

In the latter case, no sequence of purchases can distinguish  $\theta$  from  $\theta + \delta$ . Finding a single such direction suffices to show that the *full pattern* cannot be identified. It need not imply that every component is unidentified. More generally, a linear target  $\ell'\theta$  is identified precisely when  $\ell$  lies in the row span of the available design.

**Use arithmetic help without outsourcing the conclusion.** The calculator solves the observed linear equations, assigning zero to free coordinates in its elimination convention. It does not mark free coordinates, display a live rank or certify uniqueness. Its output may change dramatically when a new independent comparison is added. The fit-check button only compares submitted predictions with recorded results. Before sealing, the player must decide whether any alternative effect pattern remains possible.

## 4 The exact benchmark and scoring

Let  $A$  be the set of new pilots selected,  $A_0$  the free archived pilots and  $O_0$  the partners whose opening fees are already paid. Pilot  $e$  costs  $k_e > 0$ ; partner  $s$  has opening fee  $F_s \geq 0$ . Total new spending is

$$C(A) = \sum_{e \in A} k_e + \sum_{s \notin O_0} F_s \mathbf{1}\{\text{some } e \in A \text{ uses } s\}.$$

When the full vector can be identified, the benchmark is

$$C^* = \min_{A: \text{rank}(X_{A_0 \cup A})=d} C(A).$$

The hard budget limits purchases. The authored identifiable missions have  $B \geq C^*$ . A poor sequence can leave insufficient money to finish, but this does not make a structural impossibility statement true.

**Proposition 2.** *Enumerating subsets of unopened partners and finding a minimum-cost basis completion conditional on each subset computes  $C^*$  exactly.*

*Proof.* Fix a subset  $S$  of newly opened partners. Include the archived row span, sort eligible unpurchased rows by pilot cost, and accept each row exactly when it increases rank, stopping at rank  $d$ . Independent row sets, after quotienting out the archived span, form a linear matroid. The basis exchange property implies that this greedy procedure minimises total pilot cost for the fixed eligible set: each accepted cheapest extending row can replace a no-cheaper row in an optimal completing basis without losing span.

Add the opening fees for  $S$  and minimise over all such subsets. Every feasible study is considered under its actual set of new openings; conditional greedy costs no more than that study. Conversely, each candidate completing basis is feasible, with actual cost no larger than its charged fees. A subset charging for unused partners cannot improve on the candidate using only its actual openings. Hence the minimum is exactly the global optimum. There are at most  $2^7 = 128$  opening subsets.  $\square$

Every new pilot contributes one scalar restriction, so at least  $d - r_0$  new pilots are needed, where  $r_0 = \text{rank}(X_{A_0})$ . With positive costs, a cheapest design contains no redundant new row and uses exactly  $d - r_0$  pilots. Their *choice* and opening costs remain nontrivial. Outcomes cannot change rank or the optimal design here: adaptive outcome-dependent choices offer no identification advantage in this unrestricted linear model.

**Scoring.** A sealed numerical report earns zero unless its evidence has rank  $d$  and all effects are correct to the stated rounding tolerance of 0.005 income units. A lucky true guess under rank deficiency earns zero. A sealed impossibility report earns zero unless its entered nonzero rational direction annihilates every permitted row. Malformed or incomplete entries are returned for correction before sealing.

For a valid report, the score is

$$S = \text{round}\left(100 \frac{C^* + 1}{C_{\text{spent}} + 1}\right), \quad 1 \leq S \leq 100.$$

For a valid structural impossibility proof,  $C^* = 0$ : the known menu already supplies the proof, without purchased outcomes. The added credit makes its score well defined. Rounding can give equal scores to nearby spending levels. There are no time bonuses or penalties for free planning. The postmortem reveals the optimum and a completing design, or the structural null direction.

## 5 The four authored contracts

The following inputs are complete apart from the hidden income effects. Shares are percentages in the order  $LN, LF, HN, HF$ . In the first three missions the recruitment mixes and programme availability are:

| Partner           | $LN$ | $LF$ | $HN$ | $HF$ | Versions available |
|-------------------|------|------|------|------|--------------------|
| Town market       | 40   | 10   | 40   | 10   | $0, T, R, B$       |
| Technical college | 10   | 10   | 40   | 40   | $T, B$             |
| Bus depot         | 10   | 40   | 10   | 40   | $R, B$             |
| Outreach team     | 40   | 40   | 10   | 10   | $0, T, R$          |
| Employment office | 10   | 30   | 40   | 20   | $0, T, R, B$       |
| Village hall      | 35   | 40   | 15   | 10   | $0, T, R, B$       |
| Regional fair     | 30   | 15   | 10   | 45   | $0, T, R, B$       |

At the college, training is already promised; at the depot, transport is already promised. These partners can randomise the additional programme. Outreach can deliver either programme separately but cannot deliver the package.

An arm offering  $(T, R)$  costs  $h_s + t_s T + r_s R$ . A pilot costs the sum of its two arm costs. Fees  $F_s$  apply in missions 1, 3 and 4;  $\tilde{F}_s$  applies in mission 2.

| Partner           | $h_s$ | $t_s$ | $r_s$ | $F_s$ | $\tilde{F}_s$ |
|-------------------|-------|-------|-------|-------|---------------|
| Town market       | 2     | 3     | 2     | 9     | 25            |
| Technical college | 2     | 2     | 3     | 3     | 26            |
| Bus depot         | 2     | 4     | 1     | 4     | 15            |
| Outreach team     | 1     | 2     | 2     | 6     | 16            |
| Employment office | 1     | 2     | 1     | 16    | 31            |
| Village hall      | 3     | 5     | 2     | 5     | 5             |
| Regional fair     | 1     | 1     | 1     | 24    | 21            |

| Mission                      | $d$ | $r_0$ | New pilots <sup>†</sup> | $C^*$        | Budget |
|------------------------------|-----|-------|-------------------------|--------------|--------|
| 1. Separate the channels     | 8   | 2     | 6                       | 62           | 81     |
| 2. The cost of opening doors | 8   | 0     | 8                       | 117          | 133    |
| 3. Let every group differ    | 12  | 1     | 11                      | 103          | 110    |
| 4. Everyone is represented   | 8   | 1     | —                       | $0^\ddagger$ | 130    |

<sup>†</sup>Minimum for a numerical identification report. <sup>‡</sup>Cost of a valid impossibility proof; no numerical identifying design exists.

Mission 1 supplies market  $B = 0$  and outreach  $T = 0$  results, with both openings paid. Mission 2 supplies none. Missions 3 and 4 supply market  $B = 0$ , with its opening paid. A fresh round changes the effects. A *new contract* also changes all prices and recomputes the exact optimum and budget; its scores are tracked separately. The listed budgets and costs describe the authored prices only.

In mission 4 every version is available at every partner. Its shares, in the same partner order as above, are

$$(25, 25, 25, 25), (35, 45, 5, 15), (40, 20, 30, 10), (10, 30, 20, 40), \\ (45, 25, 25, 5), (15, 5, 45, 35), (5, 35, 15, 45).$$

Each group appears everywhere. The special relation among the shares, not absence of a demographic group, drives the failure of identification.

## 6 Worked solutions and a certificate of impossibility

*This section reveals the authored solutions.* The tables group new pilots by partner; the order of purchases is immaterial. Free archived results are retained in every design. Each listed set supplies exactly the missing independent restrictions. Other sets may tie the minimum cost.

| Partner                              | New comparisons       | Pilots cost | Opening | Total |
|--------------------------------------|-----------------------|-------------|---------|-------|
| <b>Mission 1: minimum 62 credits</b> |                       |             |         |       |
| Employment office                    | $R - 0, T - 0, B - 0$ | 12          | 16      | 28    |
| Town market                          | $R - 0$               | 6           | 0       | 6     |
| Bus depot                            | $B - R$               | 10          | 4       | 14    |
| Technical college                    | $B - T$               | 11          | 3       | 14    |

| Partner                               | New comparisons       | Pilots cost | Opening | Total |
|---------------------------------------|-----------------------|-------------|---------|-------|
| <b>Mission 2: minimum 117 credits</b> |                       |             |         |       |
| Regional fair                         | $T - 0, R - 0, B - 0$ | 10          | 21      | 31    |
| Employment office                     | $R - 0, T - 0, B - 0$ | 12          | 31      | 43    |
| Bus depot                             | $B - R$               | 10          | 15      | 25    |
| Village hall                          | $B - 0$               | 13          | 5       | 18    |

| Partner                               | New comparisons       | Pilots cost | Opening | Total |
|---------------------------------------|-----------------------|-------------|---------|-------|
| <b>Mission 3: minimum 103 credits</b> |                       |             |         |       |
| Regional fair                         | $T - 0, R - 0, B - 0$ | 10          | 24      | 34    |
| Employment office                     | $R - 0, T - 0, B - 0$ | 12          | 16      | 28    |
| Outreach team                         | $T - 0, R - 0$        | 8           | 6       | 14    |
| Town market                           | $R - 0, T - 0$        | 13          | 0       | 13    |
| Technical college                     | $B - T$               | 11          | 3       | 14    |

**Mission 4.** Every recruitment vector satisfies  $p_{LN} + p_{HF} = p_{LF} + p_{HN} = 1/2$ . Set the proposed changes to all four shared main effects to zero, and change the interactions by

$$(\delta c_{LN}, \delta c_{LF}, \delta c_{HN}, \delta c_{HF}) = (1, -1, -1, 1).$$

For any partner and any programme contrast, the change in its reported result is

$$\Delta(TR)\{p_{LN} - p_{LF} - p_{HN} + p_{HF}\} = 0.$$

The nonzero direction is therefore invisible even to the full menu. Its rank is seven, so this is the sole missing dimension. Main effects are identifiable from the full menu; the ambiguity lies in the interaction pattern. Buying more pilots cannot remove it. A partner whose mix breaks the displayed equality, or suitable group-specific reporting, could supply the missing information, but neither is available under this contract.

## 7 What the game teaches, and what it does not

**Identification is a property of the information and model together.** Randomisation removes selection within a comparison. It does not disaggregate a pooled response, separate programme components never varied independently, or justify transferring effects across partners without assumptions. A representative package trial can identify its average joint effect perfectly while leaving the requested decomposition unresolved. The model restrictions in the eight-effect missions genuinely reduce the required information; dropping them in mission 3 changes the design problem.

**Coverage is weaker than independent variation.** Mission 4 is a concrete example: every group has positive weight at every site, and every programme comparison is available. Yet the same offsetting interaction change cancels everywhere. The player can establish this before observing any income effects. Such a proof is stronger than a statement that the present budget or notebook seems insufficient.

**Efficiency and validity are separate.** A costly but identifying design supports a correct conclusion and earns a positive score. A cheap unsupported conclusion earns zero. The cost benchmark discourages collecting every available contrast, but the score never rescues a false final statement. A valid proof of impossibility after unnecessary purchases earns less than the same proof made directly from the known design.

**Checks against easy purchasing rules.** At authored prices, repeatedly buying the cheapest currently rank-increasing comparison, including any immediate opening fee, costs 63, 144 and 115 credits in missions 1–3 under the implementation’s deterministic tie-breaking. The latter two exceed their budgets of 133 and 110. Collecting the three programme-versus-neither contrasts at all four partners offering every version costs 112, 158 and 112 credits respectively, even after allowing for archived results and paid openings. Each exceeds its mission budget. These are specific shortcut checks, not evidence that every expert will find the game difficult.

**A possible teaching sequence.** Ask students what adding training to existing transport measures, and why a further comparison might add no information. Let them discuss a study before using the calculator, then compare their completed design with the benchmark. In the final mission, ask for two interaction tables agreeing on every possible result, and what new data would distinguish them. Experienced players can use the comparison-weight table and attempt optimal designs under fresh prices.

**Scope and validation.** The game concerns group mean effects, not individual treatment effects or the distribution of gains. It has no noncompliance, interference, sampling variation or model uncertainty. The unrestricted real parameter space is the public model even though hidden coefficients are generated on a fine grid to make reports readable. Removing the pooled-reporting restriction, allowing arbitrary recruitment or allowing shared arms changes the experiment menu and its benchmark. A freely designed, stratified factorial trial is a different information environment. The intended expert challenge is the interaction between heterogeneous effects, independent contrasts and fixed costs; the outcome values themselves do not create an adaptive search puzzle.

The implementation uses exact rational arithmetic for rank, fitting and certificates. Its optima were cross-checked in Python; the optimiser also matched exhaustive searches on small problems. Checks covered 160 authored and repriced solution paths, unsupported true guesses, false reports and budget limits. Actual interface handlers were exercised in a DOM simulation through 32 complete playthroughs and 200 pilot runs. This validates functions, not browser layout or classroom difficulty. The standalone HTML needs no network connection. Its answer key is inspectable, so scores serve learning rather than secure assessment.